ModelIt!

Teacher's Guide

Why Is It Dark at Night?

The Science of Day and Night Patterns

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-ESS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-ESS1-1

Use observations of the sun, moon, and stars to describe patterns that can be predicted.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students observe and describe the pattern of day and night caused by Earth's rotation.
Disciplinary Core Idea	**ESS1.A The Universe and its Stars** — Patterns of the motion of the sun, moon, and stars in the sky can be observed, described, and predicted.
Crosscutting Concept	**Patterns** — Students identify that day and night is a repeating, predictable pattern caused by Earth's rotation.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Day and night is one of the most important patterns in nature. Understanding why it happens helps us learn how our planet moves through space.

Why Day and Night Happen

The Earth is always spinning, rotating once every 24 hours. This spinning is what causes day and night. The sun stays in one place — it does not rise and set like it looks from the ground. Instead, the Earth turns so that different parts face the sun at different times. When your part of the Earth faces the sun, it is daytime. When your part rotates away from the sun, it is nighttime. This is why day and night always follow each other in a predictable pattern.

A Pattern We Can Predict

Because the Earth spins at the same steady speed, day and night follow a very reliable pattern. We know exactly when the sun will rise and set every single day. This pattern is so reliable that people have used it to tell time for thousands of years. Sunrise, daytime, sunset, nighttime — and then it starts all over again. The whole cycle takes about 24 hours, which is why we say a day is 24 hours long.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Earth's position as the external component we can set (facing sun or away) and amount of sunlight as the internal result that changes.

E **Establish Relationships**
Students discover that when your side of Earth faces the sun, sunlight is high (day), and when it faces away, sunlight is zero (night).

V **Visualize & Model**
Students build a simple model with two parts showing how Earth's position controls the amount of sunlight.

E **Evaluate Outcomes**
Students test their model by simulating both positions and comparing the results.

R **Revise & Extend**
Students reflect on how their model explains the daily pattern of day and night they experience.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Why is it bright outside right now but it will be dark tonight? Where does the sun go? Or does it go anywhere at all?"*

Watch For

Ask students to share what they think happens to the sun at night. Write ideas on the board to revisit later.

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn WHY it gets dark at night. We will learn a big word: ROTATE. It means to spin around!"*

Watch For

Have students stand up and rotate in place slowly. Say: That is what the Earth does all the time!

Slide 3: Let's Be Scientists!

3 min

Say: *"Here is our big question: Why does the sky get dark at night? Does the sun really go away, or does something else happen?"*

Watch For

Think-pair-share. Students share their ideas with a partner. Collect predictions.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Earth's Position is which way we face. Amount of Sunlight is how much light we get."*

Watch For

Hold up a ball (Earth) and a flashlight (sun). Shine the light on one side. Ask: Is the other side bright or dark?

Slide 5: How Are They Connected?

5 min

Say: *"When our side faces the sun, we get sunlight — that is DAY. When we spin away, no sunlight — that is NIGHT!"*

Watch For

Slowly rotate the ball in front of the flashlight. Students watch the lit side and dark side switch. Use sentence frame.

Slide 6: What Did We Find Out? 5 min

Say: *"The sun does NOT go away at night! The Earth spins and our side turns away from the sun. It is a pattern that happens every single day."*

Watch For

Revisit predictions from Slide 1. Ask: Were we right? What did we learn that surprised us?

Slide 7: STEM Challenge 10 min

Say: *"Now build your own model to show a younger student why it gets dark at night. Use a ball and flashlight!"*

Watch For

Distribute balls, flashlights, and stickers (to mark your location). Students build and demonstrate their models.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Earth's Position
- **INTERNAL:** Amount of Sunlight

Relationships

- **Earth's Position to Amount of Sunlight:** POSITIVE (+) — When your side of Earth faces more toward the sun, you get more sunlight. Facing the sun means daytime. Facing away means nighttime.

Key Discoveries

- The Earth spins slowly all the time, which causes day and night
- When your side of Earth faces the sun, it is daytime and the sky is bright
- When your side of Earth faces away from the sun, it is nighttime and the sky is dark
- Day and night is a pattern that repeats every single day

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The sun goes away at night and comes back in the morning"	The sun never goes away! It stays in the same spot all the time. What changes is which part of the Earth faces the sun. When your side turns away from the sun, it looks like the sun disappears, but really YOU moved, not the sun.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The moon makes it dark at night"	The moon does not cause nighttime. Nighttime happens because your part of the Earth has rotated away from the sun. The moon actually reflects sunlight — that is why it glows at night! Sometimes the moon is visible during the day too.